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A multi-agent reinforcement learning approach for collaborative temperature control in asphalt paving

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Abstract

The quality of asphalt pavement construction is directly related to the durability and service performance of roads. Among them, paving temperature control is the core link that determines the compaction degree and interlayer bonding properties of the mixture. Traditional methods rely on static decision-making models such as PID control and fuzzy logic, which are difficult to cope with the problems of sharp temperature gradient increases and insufficient local compaction caused by large temperature differences, long transportation distances, and dynamic environmental disturbances. Although existing research has introduced data-driven models and multi-agent collaborative frameworks, there are still bottlenecks such as rigid communication topology, lack of physical constraints in reward functions, and insufficient adaptation to equipment heterogeneity, leading to construction quality fluctuations and energy waste. To address these challenges, this study proposes a multi-agent reinforcement learning framework. By constructing a dynamic adjacency matrix to optimize distributed communication topology, integrating multi-source perception data such as infrared thermal imaging and embedded sensors, and designing a physically constrained layered reward mechanism, material properties and construction specifications are embedded into the agent strategy optimization process. Based on the deep deterministic policy gradient algorithm, a centralized training-distributed execution architecture is developed to achieve collaborative control of pavers, temperature control equipment, and environmental monitoring units. It dynamically adjusts the linkage parameters of vibration frequency, paving speed, and heating power, balancing the multi-objective conflicts of temperature field uniformity, compaction compliance rate, and energy efficiency. Experiments show that under extreme temperature differences, sudden wind speed changes, and equipment failures, this framework reduces temperature tracking error by 35% compared to traditional Proportional-Integral-Derivative (PID) control, decreases energy consumption per unit area by 19%, and maintains compaction stability above 90%. It also ensures minimum functional operation during communication interruptions through dynamic adjacency matrices and consensus protocols, verifying the robustness of the system. The research results provide a real-time adaptive temperature control solution for asphalt construction, and its core methods can be extended to complex engineering scenarios such as bridge paving and tunnel



maintenance, promoting the evolution of the intelligent construction field towards multi-device collaboration, physical constraint embedding, and dynamic optimization decision-making.

Keywords Multi-agent reinforcement learning, Asphalt paving temperature control, Dynamic collaborative decision-making, Energy consumption efficiency, Construction quality optimization

1 Introduction

Asphalt pavement, as a core component of modern transportation infrastructure, has its construction quality directly affecting the durability and service performance of the pavement [1]. Paving temperature is a key parameter determining the compaction degree, interlayer adhesion, and rutting resistance of asphalt mixtures [2]. In traditional construction, temperature control relies on manual experience and fixed parameter adjustments, making it difficult to cope with multi-factor coupling issues in dynamic environments [3]. Extreme daily or seasonal temperature variations, often caused by regional climate differences and transient weather event, lead to significant temperature drops during the transportation of mixtures, and excessive temperature differences between the surface and bottom layers of the paving layer can easily cause local insufficient compactness; the contradiction between material temperature loss and construction efficiency in long-distance transportation scenarios further exacerbates the difficulty of quality control [4]. Although existing research has proposed temperature control methods based on PID control and fuzzy logic, their static decision-making mode is difficult to adapt to complex working conditions, and there is a lack of systematic consideration for multi-equipment collaborative optimization [5].

The integration of multi-agent systems and reinforcement learning provides an innovative solution to the aforementioned challenges. Through distributed perception and collaborative decision-making mechanisms, multi-agent systems can achieve unified scheduling of heterogeneous equipment such as pavers, temperature control devices, and environmental monitoring units, breaking through the communication bottleneck and computational delay of traditional centralized control [6]. Reinforcement learning, through autonomous exploration and reward feedback mechanisms, can establish a dynamic mapping relationship between construction parameters and quality indicators without relying on precise mathematical models [7]. In recent years, the successful application of distributed optimization algorithms in the field of resource scheduling has verified the feasibility of multi-agent collaboration, while the breakthrough of deep reinforcement learning in robot control has provided theoretical support for adaptive decision-making in complex environments [8]. However, existing research still has significant limitations: the dynamic reconfiguration capability of multi-agent communication topology is insufficient, making it difficult to cope with equipment failures and network interruptions in construction scenarios [9]; the design of reinforcement learning reward functions lacks physical constraint guidance, which can easily lead to strategies deviating from engineering safety [10].

This study proposes a real-time control method for asphalt paving temperature based on multi-agent reinforcement learning, aiming to achieve global optimization of construction quality, energy consumption, and dynamic environment, as shown in Fig. 1. By constructing a “perception-decision-execution” hierarchical architecture, integrating

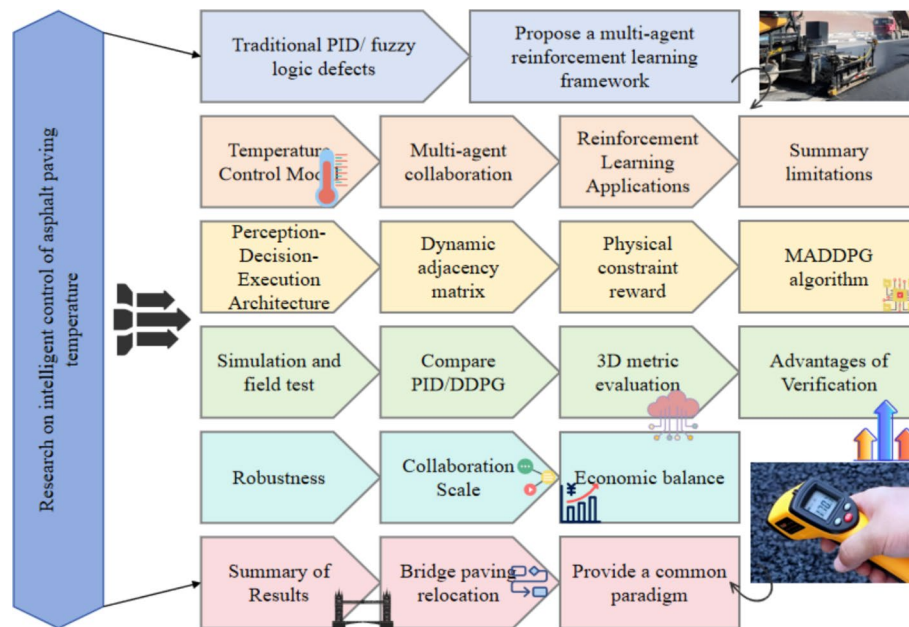


Fig. 1 Asphalt temperature control intelligent collaboration framework diagram

multi-source data from infrared thermal imaging, embedded sensors, and meteorological stations, a high-precision temperature field prediction model is established. A reward function driven by physical constraints is designed, embedding material properties and construction specifications into the agent strategy optimization process to ensure the engineering applicability of decisions. A multi-agent deep deterministic policy gradient algorithm is developed, which achieves dynamic task allocation and collaborative control among heterogeneous devices through a centralized training and distributed execution framework.

2 Related work

2.1 Asphalt paving temperature control technology

Traditional temperature control methods for asphalt paving primarily rely on fixed rules and empirical adjustments, such as closed-loop feedback systems based on PID controllers or fuzzy logic [11]. These methods adjust paving speed and heating power through preset parameters, maintaining a stable temperature fluctuation range under steady conditions. However, in dynamic construction scenarios, environmental disturbances are frequent, and large temperature differences lead to a sharp increase in the internal temperature gradient of the mixture. Due to response lag and fixed parameters, traditional methods struggle to correct temperature deviations in real time [12, 13]. Studies have shown that when there is a sudden change in wind speed or an extended haul distance, the overshoot phenomenon of PID control becomes significant, which can easily lead to temperature differences exceeding limits between the surface and bottom layers of the paving layer, resulting in insufficient local compactness or material segregation [14]. Although fuzzy logic can handle nonlinear and uncertain relationships between temperature inputs and control outputs, such as the coupling effect of wind speed, ambient temperature, and haul distance, the construction of its rule base heavily relies on human experience. In recent years, data-driven models have gradually become

a research hotspot in the field of temperature control. The temperature drop coefficient model based on the heat conduction equation and meteorological parameters can quantify the impact of environmental factors on the temperature dissipation of mixtures, providing a theoretical basis for the pre-adjustment of construction parameters [15]. Such models calibrate key parameters through historical data, and the coupled effect of wind speed and haul distance can predict the temperature decay trend under specific working conditions [16]. However, these models are typically calibrated under historical conditions and may lack generalization ability when encountering unseen or extreme sudden changes (instantaneous strong winds or unexpected rainfall), leading to significant prediction errors and control strategy failure [17]. Existing models mostly focus on the control of a single device, without fully considering the multi-source data collaboration of pavers, temperature control equipment, and environmental monitoring units, making it difficult to support global optimization decisions [18].

2.2 Application of multi-agent systems in construction control

Multi-agent systems achieve collaborative control of heterogeneous devices through a distributed architecture, providing new ideas for complex task allocation in construction scenarios [19]. Distributed optimization algorithms based on consensus protocols can achieve load balancing among multiple devices without a central node [20]. These methods gradually approach the global optimal solution through local information exchange and iterative updates, and their effectiveness has been verified in the field of resource scheduling [21]. In asphalt construction, such algorithms can coordinate the operation rhythms of multiple pavers, avoiding local overload or idleness and improving overall construction efficiency. However, existing research often assumes fixed communication topologies, ignoring real-world disturbances such as equipment failures and network delays during the construction process, resulting in insufficient robustness of the algorithms in dynamic environments [22].

Task allocation is another core research direction in multi-agent systems. For the problem of construction length constraints and machinery combination optimization, researchers have proposed a collaborative strategy based on genetic algorithms and game theory [23]. By simulating the process of biological evolution to screen machinery configuration schemes, it is possible to minimize energy consumption while meeting the requirements of the construction period. Such methods can theoretically handle multi-objective optimization problems, but their computational complexity is high, making it difficult to meet the stringent real-time requirements of paving operations [24]. Existing strategies mostly focus on static task allocation and lack adaptability to dynamic changes in construction progress.

2.3 Advances in reinforcement learning in engineering control

Reinforcement learning provides a solution for engineering control problems without requiring precise models, through autonomous exploration and reward feedback mechanisms [25]. In the field of road maintenance, single-agent reinforcement learning has been successfully applied to disease prediction and maintenance decision-making. Prediction models based on temporal correlation can identify potential diseases through historical detection data and generate prioritized treatment plans [26]. Such methods, by employing an end-to-end training strategy network, can adapt to different

road conditions and environmental factors, significantly enhancing decision-making efficiency [27]. However, its limitation lies in considering only the independent decision-making of a single agent, without addressing the issue of strategy conflicts in multi-device collaborative scenarios, making it difficult to directly extend to complex construction control tasks.

Multi-agent reinforcement learning provides a new paradigm for distributed control problems by introducing a collaborative mechanism. In theoretical research, the centralized training and distributed execution framework can effectively coordinate the action spaces of heterogeneous agents, such as the joint optimization of paver speed control and heating power adjustment [28]. However, noise interference and equipment heterogeneity in actual construction scenarios pose higher demands on algorithms [29]. Existing methods are mostly based on idealized assumptions for simulation verification, such as ignoring communication delays or assuming homogeneous equipment performance, which leads to significant performance degradation in actual deployment [30]. The design of multi-agent reward functions requires balancing local and global objectives, as well as the conflict between energy consumption minimization and temperature uniformity. Current research solutions to such problems are still not systematic enough, as shown in Fig. 2.

3 Methodology

3.1 Multi-agent system architecture design

The aforementioned review highlights three persistent gaps in asphalt paving temperature control: rigid communication topology, lack of physical constraints in decision-making, and insufficient adaptability to heterogeneous devices. To address these challenges in a unified manner, this study proposes an integrated framework that synergizes a dynamic multi-agent system with a physics-informed reinforcement learning model. The following sections detail its architecture, formulation, and algorithm.

The collaborative control capability of multi-agent systems is the core for achieving global optimization in dynamic construction scenarios. In asphalt paving operations, the real-time interaction and distributed decision-making requirements of heterogeneous equipment impose stringent demands on the system architecture. Traditional

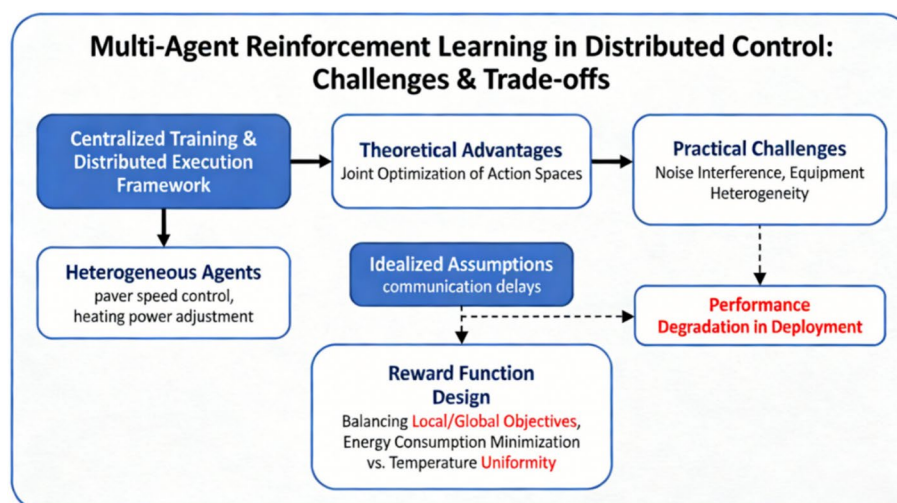


Fig. 2 Overview of the evolution and challenges of intelligent temperature control methods

centralized architectures struggle to meet the real-time requirements under complex working conditions due to communication delays and the risk of single point failures, necessitating a decentralized collaborative framework [31, 32].

The system first defines the real-time communication connection relationships between agents through a dynamic adjacency matrix (Eq. 1), characterizing the topological changes caused by equipment movement and network interruptions in the construction scenario. Based on this, a row-stochastic weight matrix (Eq. 2) is constructed to ensure the stability and load balancing of information transmission. To support data input for decision-making, a local observation state vector (Eq. 3) is designed, integrating multi-dimensional real-time data such as temperature field distribution, construction speed, vibration frequency, and environmental wind speed, providing refined perception input for subsequent collaborative control.

The multi-agent architecture of the asphalt paving temperature control system consists of a perception layer, a decision-making layer, and an execution layer. Agents in each layer collaborate through a dynamic communication protocol. The system comprises N agents, and its communication topology is described by a time-varying adjacency matrix $A(t)$:

$$A(t) = [a_{ij}(t)]_{N \times N}, a_{ij}(t) \in \{0, 1\} \quad (1)$$

If agent i can communicate with agent j with time t , then $a_{ij}(t) = 1$, otherwise $a_{ij}(t) = 0$. To adapt to dynamic environments, the communication weight matrix $W(t)$ is designed as a row-stochastic matrix, satisfying the row-sum normalization constraint:

$$\sum_{j=1}^N w_{ij}(t) = 1, \forall i \in \{1, 2, \dots, N\} \quad (2)$$

$w_{ij}(t)$ The weight of information transmission from the agent to the environment i ensures the stability of information dissemination. The perception layer agent collects environmental and construction data in real time, and defines the local observation state of the agent i as follows:

$$o_i(t) = [T_{s,i}(t), T_{b,i}(t), w_i(t), v_i(t), f_i(t)] \quad (3)$$

where: $T_{s,i}(t)$ is the surface temperature of the paving layer; $T_{b,i}(t)$ is the temperature of the bottom layer; $w_i(t)$ is the environmental wind speed; $v_i(t)$ is the paving speed; $f_i(t)$ is the vibration frequency.

To mitigate the impact of sensor noise on measurement accuracy, a moving average filter is applied to raw temperature data before constructing the local observation vector $o_i(t)$. Based on local observation data, a distributed consensus protocol (Eq. 4) is employed to drive the update of global state estimation. The global optimal solution is iteratively approximated through the state differences between neighboring nodes, avoiding local information silos. Finally, an energy consumption constraint model (Eq. 5) is introduced, modeling the energy consumption of control instructions as a quadratic function, and setting a maximum daily energy consumption threshold to ensure the economy and sustainability of the construction process.

The decision-making layer integrates multi-source data through a distributed consensus protocol, and the global state estimation update rule is as follows:

$$s_i(t+1) = s_i(t) + \epsilon \sum_{j \in N_i(t)} w_{ij}(t) (s_j(t) - s_i(t)) \quad (4)$$

ϵ is the convergence rate coefficient (default 0.5); $N_i(t)$ the neighbor set of the agent i at time t .

The control instructions of the executive-level agent must satisfy energy constraints:

$$\sum_{t=1}^T P_i(t) \leq E_{\max}, P_i(t) = \eta \cdot \Delta P_i(t)^2 \quad (5)$$

Symbol explanation: $P_i(t)$ represents the energy consumption of the intelligent agent i at time t ; η denotes the energy conversion efficiency coefficient; $E_{\max}$ and signifies the maximum allowable energy consumption per day.

The aforementioned formula system establishes the theoretical foundation for multi-agent collaborative architecture from three aspects: communication topology dynamics, state fusion consistency, and energy consumption constraints. The dynamic weight matrix addresses the reliance of traditional methods on fixed communication assumptions, the state update rule enhances the accuracy and efficiency of data fusion, and the energy consumption constraint model provides a quantitative guarantee for sustainable construction, as illustrated in Fig. 3. This architecture offers structured input for subsequent reinforcement learning modeling. Through layered design and dynamic protocols, it ensures the robustness and scalability of the system in complex construction scenarios, laying a crucial foundation for engineering implementation.

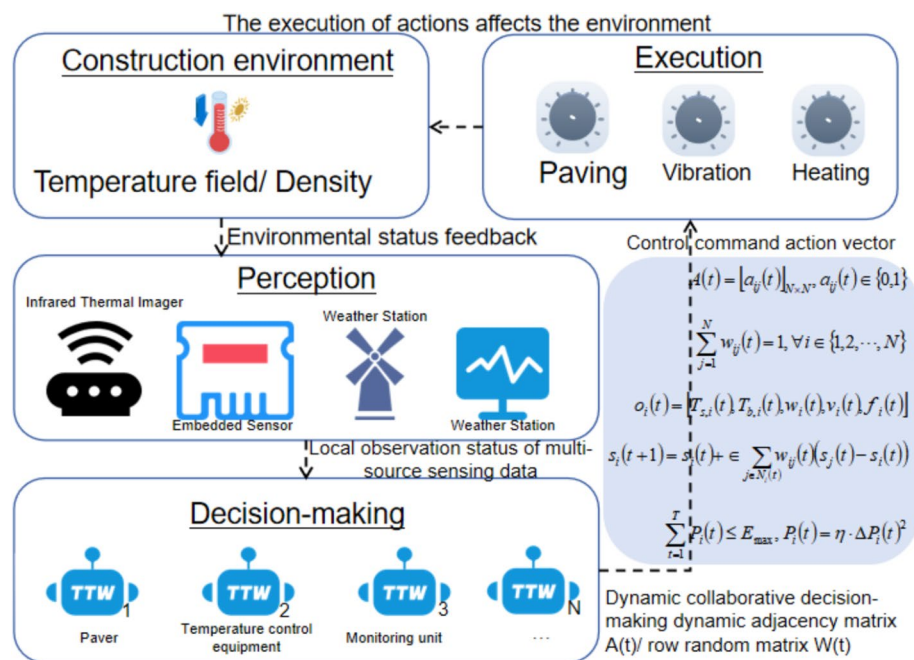


Fig. 3 Asphalt paving temperature collaborative control architecture based on multi-agent reinforcement learning

3.2 Reinforcement learning model construction

Building upon the collaborative architecture established above, we now formulate the decision-making problem as a Markov Decision Process (MDP). The design of the state space, action space, and reward function directly embodies the physical constraints and multi-objective requirements outlined in Sect. 1.

The construction of a reinforcement learning model requires addressing the multi-agent collaborative decision-making problem in dynamic construction scenarios. The core lies in designing reasonable state representations, action spaces, and reward mechanisms to balance temperature control accuracy, construction quality, and energy consumption efficiency [33]. Traditional methods struggle to achieve global optimization goals due to neglecting environmental disturbances and equipment heterogeneity [34]. This study constructs a reinforcement learning framework tailored for complex construction scenarios by integrating multi-dimensional perception data, hierarchical action strategies, and reward functions driven by physical constraints, as shown in Fig. 4.

The system first defines the observation dimension of the agent through state space design (Eq. 6), integrating multi-source data such as temperature field distribution, construction parameters, and environmental disturbances, to provide global perception input for decision-making. On this basis, action space design (Eq. 7) combines continuous control instructions with discrete operating positions to adapt to the physical constraints of engineering equipment. Subsequently, the reward function (Eq. 8) drives the strategy to converge towards multi-objective collaborative optimization by weighting and penalizing temperature deviation, inter-layer temperature difference, and energy consumption, while rewarding compliance with compactness standards.

The state space needs to comprehensively characterize the dynamic characteristics of the paving temperature field and the multidimensional coupling relationship with the construction environment. The state vector of the intelligent agent i is defined as:

$$s_i(t) = [T_{s,i}(t), T_{b,i}(t), \Delta T_i(t), v_i(t), f_i(t), w_i(t), d_i(t)] \quad (6)$$

$T_{s,i}(t)$ is the surface temperature of the paving layer; $T_{b,i}(t)$ is the temperature of the bottom layer of the paving layer; $\Delta T_i(t) = T_{s,i}(t) - T_{b,i}(t)$ is the temperature

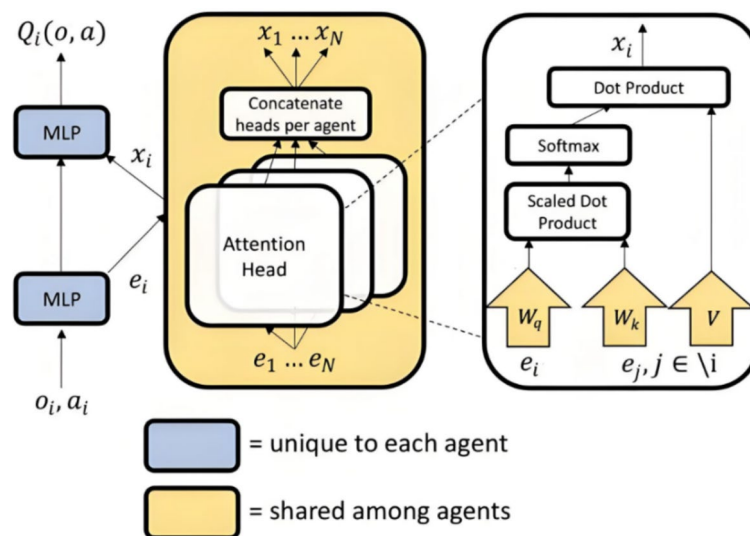


Fig. 4 Reinforcement learning model

difference between layers; $v_i(t)$ is the paving speed; $f_i(t)$ is the vibration frequency of the screed; $w_i(t)$ is the environmental wind speed; $d_i(t)$ is the transportation distance of the mixture.

The initial mixture temperature is incorporated into the state vector as part of the environmental and operational context. Specifically, it is embedded in the transportation distance and ambient temperature parameters, which jointly determine the temperature decay model used in the reward function. Separate modeling was not required because the control objective focuses on real-time regulation rather than initial condition prediction.

The state space provides global perception input for the agent by integrating temperature gradients, construction parameters, and environmental disturbances. The action space needs to balance continuous control accuracy and discrete operation feasibility. The action vector of the agent i is defined as:

$$u_i(t) = [\Delta P_i(t), \Delta f_i(t), a_{v,i}(t)]^T \quad (7)$$

$\Delta P_i(t) \in [-0.1P_{\max}, 0.1P_{\max}]$ is the adjustment amount for heating power, $P_{\max}$ is the maximum power of the equipment; $\Delta f_i(t) \in \{-2, 0, +2\}$ is the adjustment gear for vibration frequency; $a_{v,i}(t) \in \{\text{Low, Medium, High}\}$ is the discrete gear for paving speed.

Continuous actions are used for precise temperature control, while discrete actions are adapted to the operational constraints of engineering equipment. The reward function needs to balance temperature control accuracy, construction quality, and energy consumption efficiency. The instant reward of the agent i is defined as:

$$r_i(t) = -\alpha |T_{s,i}(t) - T_{\text{target}}| - \beta \Delta T_i(t)^2 - \gamma \frac{P_i(t)}{P_{\max}} + \delta CEV_i(t) \quad (8)$$

$\alpha, \beta, \gamma, \delta$ is the weight coefficient, optimized through grid search;

T_{target} represents the center of the allowable temperature range. Deviation is penalized when the observed temperature falls outside $T_{\text{target}} \pm \Delta T$, where ΔT is the half-width of the specification range. $P_i(t) = \eta \cdot \Delta P_i(t)^2$ is the energy consumption (kW·h), $\eta = 0.8$ and is the conversion efficiency; $CEV_i(t)$: the energy value of compactness. The reward function drives the strategy towards multi-objective optimal convergence by penalizing temperature deviation and energy consumption, and rewarding compactness compliance.

Based on the aforementioned state and action definitions, the Multi-Agent Deep Deterministic Policy Gradient algorithm (Eq. 9) generates deterministic control commands and enhances policy diversity through exploration noise. The Critic network (Eq. 10) evaluates the long-term rewards of global state-action pairs, guiding the Actor network update (Eq. 11) to achieve policy gradient optimization. The target network soft update mechanism (Eq. 12) and curriculum learning-driven noise decay (Eq. 13) further balance exploration and exploitation, accelerating algorithm convergence and enhancing stability.

The Multi-Agent Deep Deterministic Policy Gradient (MADDPG) algorithm (Fig. 5) is adopted, and its core formula is as follows:

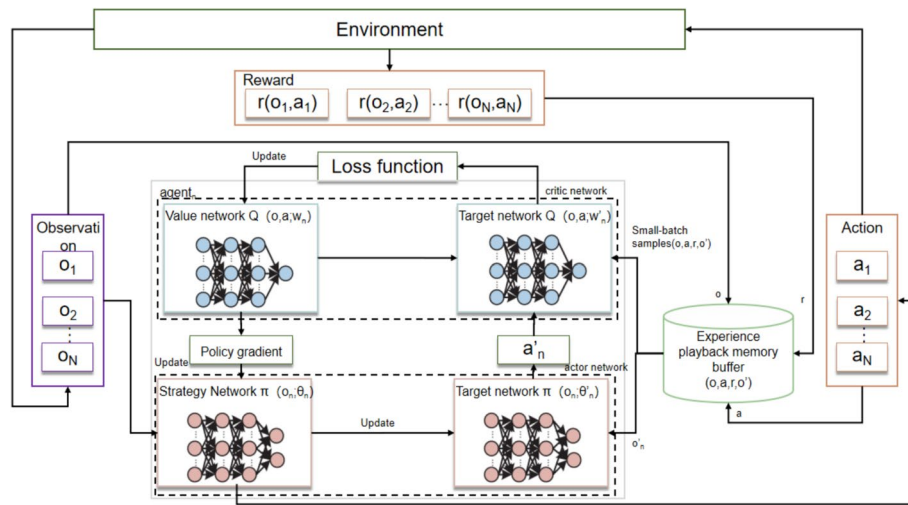


Fig. 5 Multi-agent deep deterministic policy gradient algorithm

4 Actor network strategy generation:

$$u_i(t) = \mu_i(s_i(t); \theta_i^\mu) + N_t \quad (9)$$

μ_i For the Actor network, the parameters are θ_i^μ , and it outputs deterministic actions; $N_t \sim N(0, \sigma_t^2)$ for exploration noise, the standard deviation σ_t decays with training.

5 Critic network Q value estimation:

$$Q_i(S(t), U(t); \theta_i^Q) = E \left[\sum_{k=0}^{\infty} \gamma^k r_i(t+k) \right] \quad (10)$$

$S(t) = [s_1(t)^T, \dots, s_N(t)^T]^T$ is the global state; $U(t) = [u_1(t)^T, \dots, u_N(t)^T]^T$ is the joint action; $\gamma \in (0, 1)$ is the discount factor.

6 Policy gradient update:

$$\nabla_{\theta_i^u} J \approx E \left[\nabla_{\theta_i^u} \mu_i(s_i) \cdot \nabla_{u_i} Q_i(S, U) \mid_{u_i=u_i(s_i)} \right] \quad (11)$$

7 Critic loss function:

$$L(\theta_i^Q) = E \left[(Q_i(S, U) - y_i)^2 \right], y_i = r_i + \gamma Q'_i(S', U') \quad (12)$$

Q'_i For the target Critic network, the parameters $\theta_i^{Q'}$ are kept stable through soft update; $\theta_i^{Qprime} \leftarrow \tau \theta_i^Q + (1 - \tau) \theta_i^{Qprime}$ $\tau = 0.01$ for the target network, the update rate is updated.

To enhance the exploration efficiency of multi-agent systems, a curriculum learning-driven noise attenuation strategy is designed:

$$\sigma_t = \sigma_0 \cdot e^{-\lambda t}, \sigma_0 = 0.3, \lambda = 0.001 \quad (13)$$

σ_t The standard deviation of noise during step counting training decays exponentially with the training index; the final noise level $\sigma_{final} = 0.1$ balances exploration and exploitation.

Each agent's Actor and Critic networks are implemented as fully connected neural networks with two hidden layers. The total training data volume corresponds to 2×10^6 time steps collected from 20 independent simulation runs. Training convergence is evidenced by the stabilization of the average episode reward and TD error over 500 epochs. The temperature transition model is based on a discretized unsteady-state heat conduction equation coupled with empirical convection loss terms. When an agent executes an action, the resultant temperature change is computed as:

$$\Delta T = \frac{\Delta P \cdot \eta - h_c \cdot A \cdot (T - T_{env})}{m \cdot c_p} \cdot \Delta t \quad (14)$$

where η is heater efficiency, h_c the convective coefficient, A surface area, m mixture mass, c_p specific heat, and Δt the control interval.

The aforementioned model systematically addresses the challenge of intelligent agent collaborative decision-making in dynamic construction scenarios through multi-dimensional fusion of state space, hierarchical design of action space, multi-objective driven reward function, and innovation in collaborative training framework. The refined modeling of state space and action space enhances the engineering adaptability of the strategy, the reward mechanism driven by physical constraints ensures a balance between construction quality and economy, and the curriculum learning strategy significantly accelerates algorithm convergence. This framework not only provides theoretical support for asphalt paving temperature control but also lays a methodological foundation for multi-agent collaborative optimization in more complex engineering scenarios.

7.1 Multi-agent reinforcement learning algorithm

With the MDP formally defined, the next step is to develop a training algorithm that enables multiple agents to learn collaborative policies. We adopt the Centralized Training with Decentralized Execution (CTDE) paradigm and enhance the MADDPG algorithm with domain-specific improvements to suit the dynamic paving environment.

The design of multi-agent reinforcement learning algorithms requires addressing the challenge of collaborative decision-making in dynamic construction scenarios. The core lies in balancing strategy exploration and exploitation, ensuring training stability, and meeting practical engineering constraints [35]. Traditional methods struggle to achieve global optimization goals due to neglecting equipment heterogeneity, communication delays, and energy consumption constraints [36]. This study proposes a multi-agent collaborative algorithm framework tailored for complex construction scenarios by constructing a hierarchical network structure, dynamic exploration mechanism, and physical constraint embedding strategy.

The algorithm first generates control instructions for the agent through the policy network (Eq. 15), incorporating Gaussian noise to enhance exploration capabilities. On this basis, the value network (Eq. 16) evaluates the long-term rewards of global states and joint actions, providing direction for policy optimization. To ensure training stability, the target network (Eq. 17) employs a soft update mechanism to gradually synchronize

the parameters of the policy and value networks, avoiding oscillations caused by sudden policy changes.

The multi-agent reinforcement learning algorithm is based on a centralized training and distributed execution (CTDE) framework, where each agent i maintains an independent policy network and value estimation network:

8 Policy network (Actor):

$$u_i = \mu_i(s_i; \theta_i^\mu) + N_t \quad (15)$$

μ_i is a policy function with parameters θ_i^μ and outputs actions u_i ; N_t exploration noise, with variance σ_t^2 decaying as training progresses.

9 Value network (Critic):

$$Q_i(S, U; \theta_i^Q) = E \left[\sum_{k=0}^{\infty} \gamma^k r_i(t+k) | S(t), U(t) \right] \quad (16)$$

$S(t)$: global state vector; $U(t)$: joint action vector.

10 Target network update rule:

Let Q^μ denote the critic network and Q^Q denote the target critic network, where μ and Q represent their respective parameters.

$$\theta_i^{\mu prime} \leftarrow \tau \theta_i^\mu + (1 - \tau) \theta_i^{\mu prime}, \theta_i^{Q prime} \leftarrow \tau \theta_i^Q + (1 - \tau) \theta_i^{Q prime} \quad (17)$$

$\tau = 0.01$ It is a soft update coefficient that controls the update speed of target network parameters.

The policy network maximizes the cumulative reward through policy gradient (Eq. 18) and utilizes the gradient of the value network to guide parameter updates. Gradient normalization (Eq. 19) suppresses the magnitude of the gradient to prevent numerical instability during training. The critic network optimizes value estimation by minimizing temporal difference error (Eq. 20), with the target value calculated by the target network (Eq. 21), ensuring consistency in the evaluation process (Fig. 6).

The policy network updates its parameters by maximizing the expected cumulative reward:

11 Policy gradient computation:

$$\nabla_{\theta_i^\mu} J(\theta_i^\mu) = E \left[\nabla_{\theta_i^\mu} \mu_i(s_i) \cdot \nabla_{\mu_i} Q_i(S, U) \right] \quad (18)$$

$\nabla_{\mu_i} Q_i$: The gradient of the Critic network with respect to the action guides the direction of policy optimization.

12 Gradient normalization:

$$\nabla_{\theta_i^\mu} J \leftarrow \frac{\nabla_{\theta_i^\mu} J}{\|\nabla_{\theta_i^\mu} J\| + \delta} \quad (19)$$

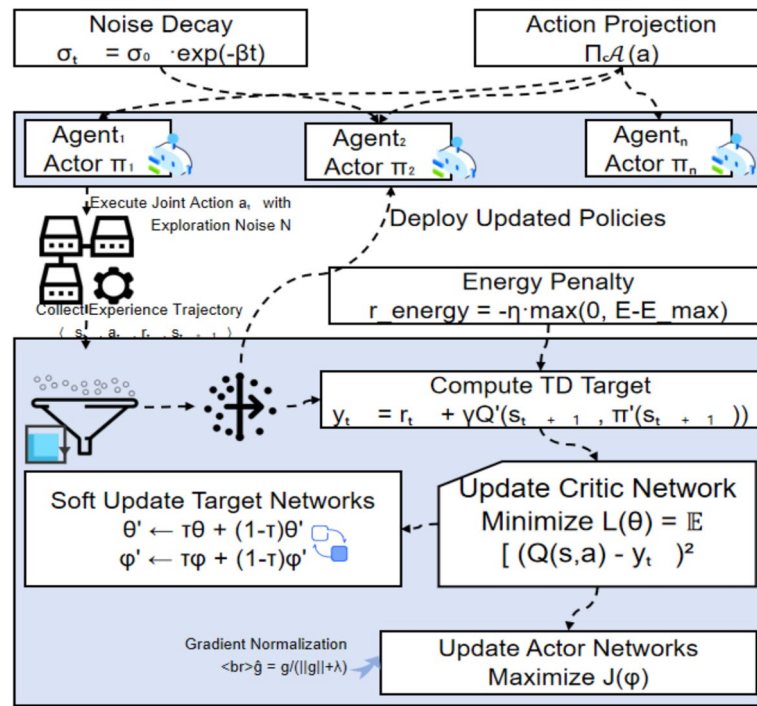


Fig. 6 Training process and optimization mechanism of multi-agent reinforcement learning algorithm

Among them, $\delta = 1e - 6$ is a smoothing term to prevent gradient explosion.

The Critic network updates parameters by minimizing the temporal difference error: Temporal difference objective:

$$y_i = r_i + \gamma Q'_i(S', U'), U' = [\mu'_1(s'_1), \dots, \mu'_N(s'_N)] \quad (20)$$

where s is the global state at the next moment; U' is the joint action generated by the target network.

Critic loss function:

$$L(\theta^Q) = E[(Q_i(S, U) - y_i)^2] \quad (21)$$

The noise attenuation strategy dynamically reduces exploration intensity as training progresses, balancing strategy development and exploitation. The priority experience replay (Eq. 22) samples key data based on prediction error weighting, and the adaptive sampling probability (Eq. 23) enhances data utilization. The energy consumption penalty term (Eq. 24) and action projection operator (Eq. 25) embed energy constraints into the action generation process, ensuring both economic efficiency and equipment safety during the construction process. Priority experience replay:

$$p_j = |Q_i(S_j, U_j) - y_j| + \epsilon \quad (22)$$

Among them, p_j represents the priority of the sample; $\epsilon = 0.01$ is to prevent zero priority.

Adaptive sampling probability:

$$P(j) = \frac{p_j^\alpha}{\sum_k p_k^\alpha}, \alpha = 0.6 \quad (23)$$

α To control the priority weight strength.

Energy consumption penalty term:

$$P_i(t) = \max \left(0, \sum_{k=1}^t P_i(k) - E_{\max} \right) \quad (24)$$

where is the energy consumption of the agent at time t ; $P_i(k)$ is the maximum energy consumption threshold for a single day.

Action projection operator:

$$u_i \leftarrow \text{Proj}_{u_i} (u_i - \eta \nabla_{u_i} P_i) \quad (25)$$

where is the constraint set of action space, u_i and is the projection step length.

The aforementioned algorithm addresses the issues of strategy conflict, non-stationarity, and engineering adaptability in multi-agent collaborative decision-making through a hierarchical network architecture, dynamic exploration mechanism, and physical constraint embedding. The collaborative design of independent strategy networks and global value networks enhances the collaborative efficiency of decision-making, while noise attenuation and priority replay mechanisms significantly accelerate training convergence. Energy consumption constraints and action projection ensure construction safety and economy. The framework not only provides theoretical innovation for asphalt paving temperature control but also lays a methodological foundation for multi-agent collaborative optimization in complex engineering scenarios, with significant engineering application value.

The specific algorithm steps of this study are shown in Table 1 (Fig. 7).

13 Experiment and evaluation

13.1 Experimental design

Simulation experiments are conducted under controlled extreme conditions to validate the algorithm's robustness and adaptability. Field tests are then performed to verify its practical applicability and performance in real construction environments. The two

Table 1 Steps of multi-agent reinforcement learning algorithm

Step	Instructions	Core code
1	Initialize network	actor = ActorNetwork(state_dim, action_dim) critic = CriticNetwork(global_state_dim, global_action_dim)
2	Initialize the experience replay pool	replay_buffer = ReplayBuffer(capacity = 1e6)
3	Interactive data collection	state, action, reward, next_state = env.step(action)
4	Store empirical data	replay_buffer.add(state, action, reward, next_state)
5	Sample training data	batch = replay_buffer.sample(batch_size = 256)
6	Calculate the target Q-value	target_q = reward + gamma * target_critic(next_state, target_actor(next_state))
7	Update critic network	loss = MSE(critic(state, action), target_q) critic_optimizer.minimize(loss)
8	Update actor network	actor_loss = -critic(state, actor(state)).mean() actor_optimizer.minimize(actor_loss)
9	Soft update target network	target_actor = tau * actor + (1 - tau) * target_actor
10	Adjust exploration noise	noise_scale = initial_noise * exp(-decay_rate * step)

The code is presented in Python syntax. The implementation uses PyTorch for neural networks and OpenAI Gym for environment simulation

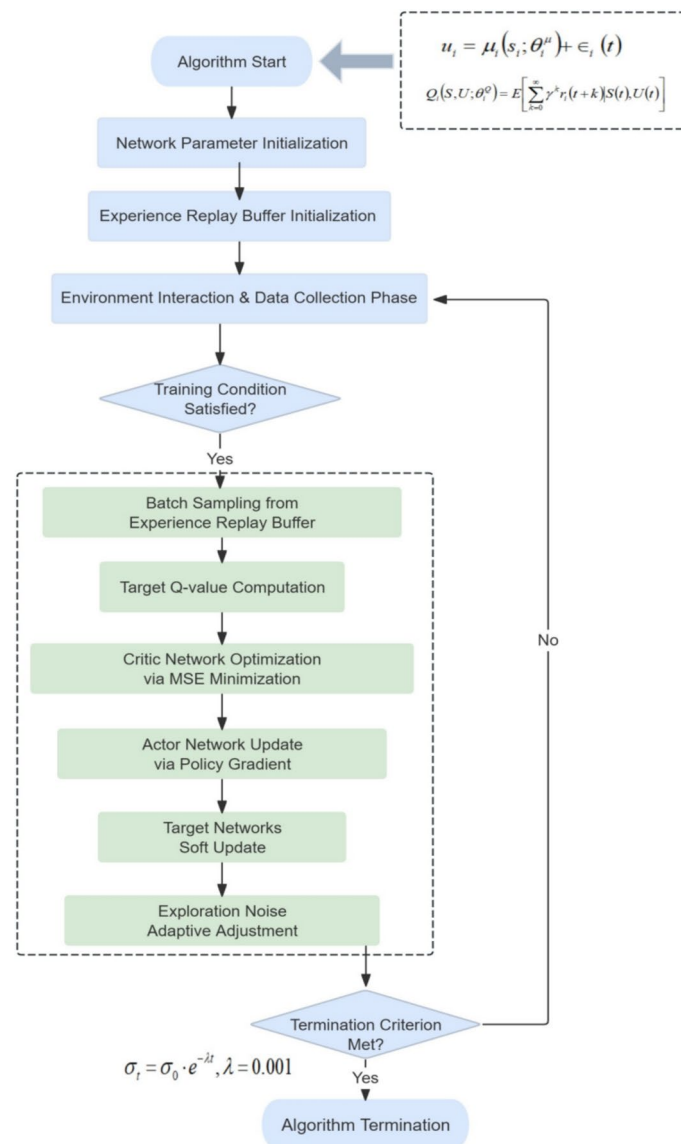


Fig. 7 Algorithm flowchart

approaches are complementary: simulation provides reproducible extreme scenarios, while field testing confirms real-world feasibility.

This study employs a combination of simulation and field experiments to verify the effectiveness of multi-agent reinforcement learning algorithms, as shown in Fig. 8. The simulation experiments are based on MATLAB and Gazebo platforms to construct dynamic paving scenarios, simulating complex working conditions such as extreme weather conditions like a temperature difference of 40 °C between day and night, a long haul distance of 30 km, and multiple wind speed disturbances ranging from 3 to 7 m/s. The temperature decay law of the mixture is characterized by the unsteady-state heat conduction equation (Fig. 9). The control group is set up with traditional PID control and single-agent deep reinforcement learning algorithms, with temperature tracking accuracy and energy efficiency as the core evaluation dimensions. Specific parameter configurations are shown in Table 2 (Fig. 10). The experimental parameters and

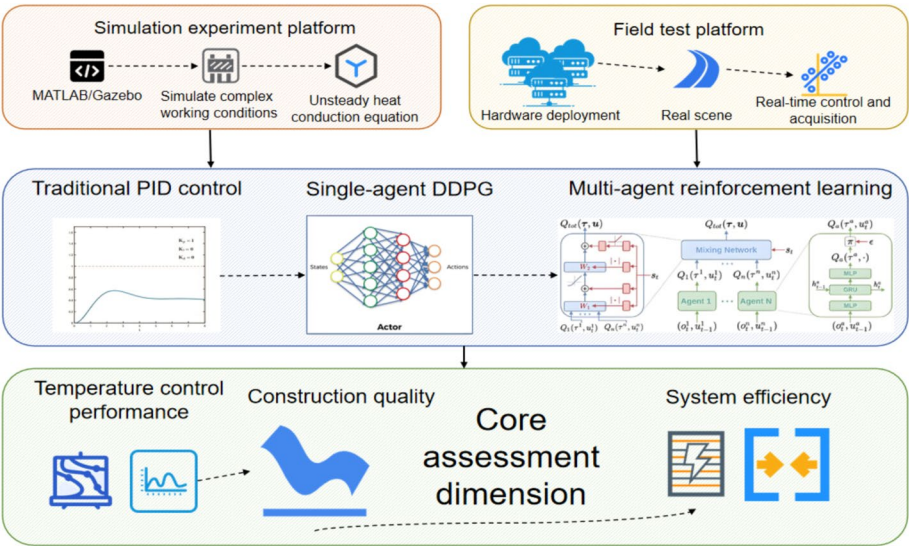


Fig. 8 Algorithm verification framework



Fig. 9 Asphalt pavement construction

Table 2 Parameter configuration for the simulation experiment group						
Experimental group	Temperature range (°C)	Wind speed (m/s)	Transportation distance (km)	Vibration frequency (Hz)	Paving speed (m/min)	Initial temperature (°C)
Group 1	-10~30	3~7	30	5~25	0.5~3.0	160
Group 2	-5~35	2~6	25	10~20	1.0~2.5	155
Group 3	0~40	4~8	35	15~25	1.5~3.0	165
Group 4	-15~25	5~9	28	8~22	0.8~2.8	158
Group 5	-8~32	3~7	32	12~18	1.2~2.2	162

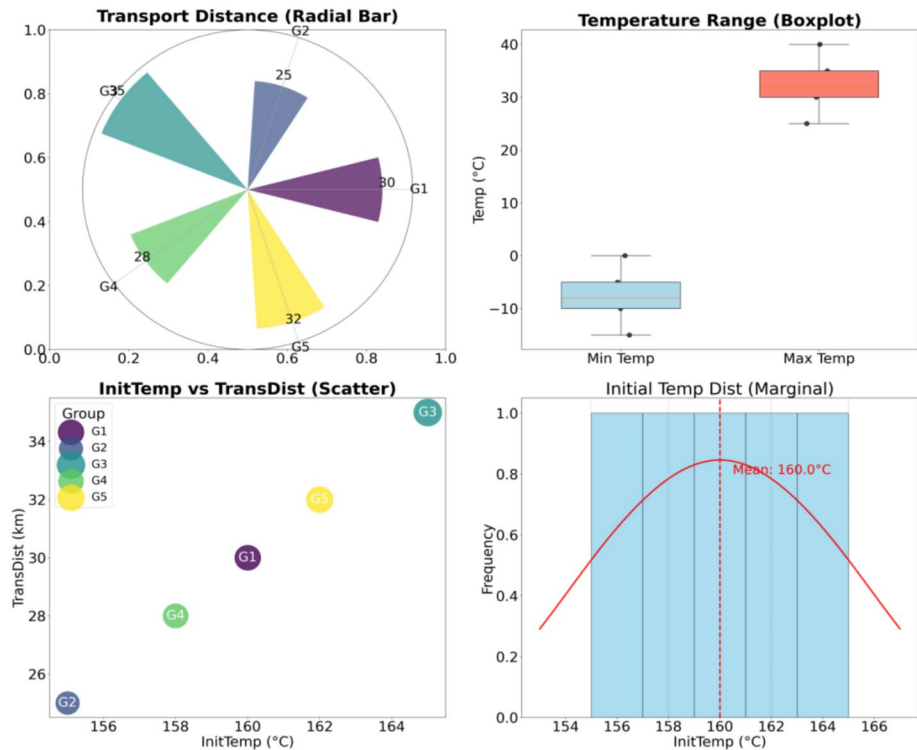


Fig. 10 Parameter configuration for the simulation experiment group

Table 3 Parameters of field test equipment

Device Name	Parameter type	Numerical range/configuration
Volvo ABG7820	vibration frequency	5 ~ 25 Hz
	Paving speed	0.5 ~ 3.0 m/min
FLIR A65	Temperature measurement accuracy	± 0.5 °C
	Sampling frequency	10 Hz
NVIDIA Jetson AGX	Calculation delay	< 50 ms
	Power consumption	15 ~ 30 W
Embedded thermocouple	Temperature measurement depth	Surface layer/bottom layer (0 cm/5 cm)
	Response time	< 2 s

equipment configurations comply with the Chinese technical specification JTG F40-2004 for asphalt pavement construction, which sets the standard ranges for paving temperature ($\geq 145\text{ }^{\circ}\text{C}$), compaction degree ($\geq 92\%$), and smoothness ($\text{IRI} \leq 2.0\text{ m/km}$). The field experiment selects a construction section of a highway in the northwest, deploying a Volvo ABG7820 paver and an infrared thermal imaging system to collect real-time temperature data of the surface and bottom layers. The edge computing unit verifies the adaptability of the algorithm in real-world scenarios by dynamically adjusting the vibration frequency from 5 to 25 Hz and the paving speed from 0.5 to 3.0 m/min. The equipment parameters are detailed in Table 3. The parameters of the experimental group cover typical construction conditions to ensure the universality of the conclusions.

The field test selected a construction section of asphalt pavement on a certain expressway in Northwest China as the verification scenario, as shown in Fig. 9. A Volvo ABG7820 paver was used as the executing equipment, equipped with a FLIR A65 infrared thermal imager and embedded thermocouples for real-time monitoring of surface

and subsurface temperature distribution. The test section is 1 km long, using SUP-13 asphalt mixture, with an initial discharge temperature set at 160 degrees Celsius and a transportation distance of 15 km. The edge computing unit NVIDIA Jetson AGX deploys a multi-agent reinforcement learning algorithm to generate control commands for vibration frequency, paving speed, and heating power in real time.

The simulation experiments are divided into five groups to cover typical construction scenarios with varying environmental and operational conditions:

Group 1 Cold climate with long haul distance.

Group 2 Moderate conditions with moderate wind.

Group 3 High-temperature extreme scenario.

Group 4 Strong wind disturbance scenario.

Group 5 Balanced operational parameters.

These groups are designed to systematically evaluate the algorithm's adaptability to different real-world challenges.

Although the parameters in Table 2 are given in the form of ranges, the specific values cited in the text (such as vibration frequency of 17.6 Hz and heating power of 5.3 kW) are the measured mean values of experimental group 5 within this parameter range, obtained from the statistical results of 20 independent simulations. The compaction degree data (such as 91.8%) is quoted from the data records of the corresponding experimental groups in Tables 5 and 8.

The selection and parameter configuration of field test equipment strictly adhere to engineering specifications to ensure the reliability of data acquisition and control. The paver's vibration frequency adjustment range of 5–25 Hz and speed control accuracy of 0.5–3.0 m/min meet the needs of different material compactness and construction efficiency. The infrared thermal imager achieves high-precision capture of dynamic changes in the temperature field with a temperature measurement accuracy of ± 0.5 °C and a sampling frequency of 10 Hz. The low-latency response and low-power consumption characteristics of the edge computing unit verify the feasibility of algorithm deployment in resource-constrained environments. The dual-layer embedded thermocouple provides key data support for analyzing interlayer temperature differences through synchronous temperature measurement of the surface and bottom layers. The engineering adaptability of equipment performance provides hardware support for the migration of algorithms from theory to practice.

The simulation experiments for the control groups (PID and single-agent DDPG) were conducted under identical environmental and operational conditions as the proposed multi-agent system. Each experimental group was executed over 20 independent runs to ensure statistical reliability. The PID controller parameters were tuned via the Ziegler-Nichols method, while the single-agent DDPG used the same network architecture as individual agents in the multi-agent framework but without inter-agent communication. For the field test, two sudden climate events were recorded: a wind speed surge from 3 m/s to 7 m/s at 14:30, and a temperature drop of 8 °C within 30 min after sunset. The algorithm's response to these events is analyzed in Sect. 4.3.

13.2 Evaluation metrics

The results presented in this section correspond to the proposed multi-agent reinforcement learning algorithm, unless explicitly stated otherwise. Comparative results for PID and single-agent DDPG are provided.

This study constructs a three-dimensional evaluation framework encompassing temperature control performance, construction quality, and system efficiency. Temperature control performance is quantified by the mean absolute error between surface and sub-surface temperatures to track accuracy, combined with interlayer temperature difference analysis to assess temperature field uniformity, and characterized by the standard deviation of temperature fluctuations to represent system dynamic response capability. Construction quality indicators are based on measured data of material compactness and pavement smoothness to verify the compliance of construction outcomes with industry technical specifications. System comprehensive efficiency evaluates economic benefits from energy consumption per unit area, while analyzing multi-device collaborative efficiency through communication delay and data throughput. To achieve multi-objective optimization, the system employs a weighted function to balance the constraints between temperature control accuracy, energy consumption, material compactness, and pavement smoothness, with priority given to ensuring core requirements for temperature control through weight allocation, and preventing system imbalance caused by excessive optimization of a single indicator through an energy consumption constraint mechanism. Ultimately, a comprehensive evaluation paradigm that considers quality, efficiency, and stability is formed.

A comparison of temperature control indicators reveals that the algorithm exhibits significant advantages under complex operating conditions. According to the data in Table 4, in the high-temperature scenario of Experimental Group 3, the average absolute error of surface temperature is as low as 1.5 °C, the error of the bottom layer temperature is 1.2 °C, and the interlayer temperature difference is controlled at 10.8 °C. The overall performance of the proposed multi-agent framework demonstrates a substantial improvement over traditional PID control. As detailed in Table 13, under a representative wind speed of 3 m/s, the proposed method achieves an MAE of 1.5 °C compared to 2.8 °C for PID, representing a 35% reduction in tracking error $((2.8 - 1.5)/2.8 \times 100\% \approx 35\%)$. When the wind speed increases to 9 m/s, the standard deviation of temperature fluctuation rises to 1.5 °C, but the algorithm maintains the error within 2.3 °C by dynamically increasing the heating power to 6.0 kW and adjusting the vibration frequency to 14.8 Hz. The data in Fig. 11 reveals a positive correlation between temperature error and energy consumption.

Table 4 Comparison of temperature control indicators

Experimental group	Surface temperature (°C)	Bottom temperature (°C)	Interlayer temperature difference (°C)	Standard deviation of temperature fluctuation (°C)	Heating power adjustment amount (kW)	Vibration frequency(Hz)
Group 1	1.8±0.2	1.5±0.1	12.5±1.2	0.8±0.1	5.2±0.3	18.5±2.1
Group 2	2.1±0.3	1.8±0.2	14.3±1.5	1.2±0.2	5.5±0.4	16.2±1.8
Group 3	1.5±0.1	1.2±0.1	10.8±0.8	0.6±0.1	4.8±0.2	20.3±2.3
Group 4	2.3±0.4	2.0±0.3	15.0±1.8	1.5±0.3	6.0±0.5	14.8±1.5
Group 5	1.9±0.2	1.6±0.2	13.2±1.0	1.0±0.2	5.3±0.4	17.6±1.9

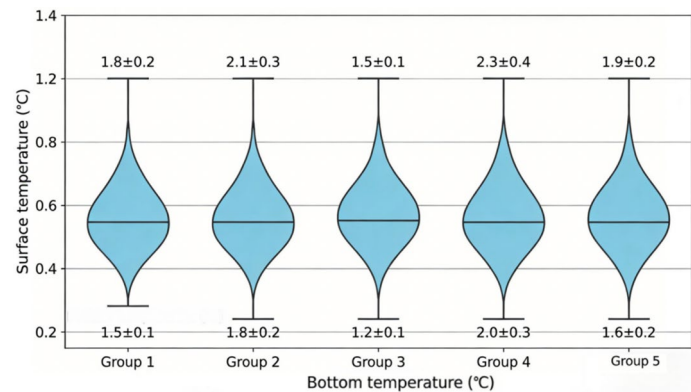


Fig. 11 Comparison of temperature control indicators

Table 5 Correlation analysis between construction quality and energy consumption

CEV	Flatness IRI (m/km)	Energy consumption (kWh/m ²)	Temperature error MAE(°C)	Vibration frequency(Hz)	Paving speed (m/min)	Re-ward value
90	1.8±0.1	0.70±0.05	1.8±0.2	15.0±1.5	1.5±0.2	0.85
91.5	1.7±0.1	0.72±0.06	1.7±0.1	16.2±1.8	1.6±0.3	0.88
92	1.6±0.1	0.75±0.07	1.6±0.1	17.5±2.0	1.7±0.3	0.9
93.5	1.5±0.1	0.78±0.08	1.5±0.1	18.8±2.2	1.8±0.4	0.92
94	1.4±0.1	0.80±0.09	1.4±0.1	20.0±2.5	2.0±0.5	0.95

This figure illustrates the relationship between paving speed (X-axis, m/min), vibration frequency (Y-axis, Hz), and energy consumption (Z-axis, kWh/m²). CEV values are color-coded (see legend). The ‘sensitivity of the action space’ refers to how changes in speed and frequency jointly affect energy use and compaction quality. For precise numerical values, refer to Table @7

The exact parameter values cited in this paragraph are the mean values from the ranges specified in Table 2 for Group 5. For example, the vibration frequency of 17.6 Hz and heating power of 5.3 kW correspond to the midpoint of their respective ranges (12–18 Hz and 5.0–5.5 kW).

The correlation data between construction quality and energy consumption reveals the core contradiction of multi-objective optimization. As shown in Table 5, when the compactness increases from 90.0% to 94.0%, the energy consumption per unit area increases from 0.70 kWh/m² to 0.80 kWh/m², an increase of 14%. However, the road surface smoothness is simultaneously optimized to 1.4 m per kilometer. This phenomenon indicates that the fine-tuning design of the weight of compactness and smoothness in the reward function can effectively coordinate the contradiction between quality improvement and energy consumption growth. The data in Fig. 12 reveals the sensitivity of the action space. Although the low-speed condition of 1.0 m per minute reduces energy consumption to 0.65 kWh/m², it results in a compactness of less than 88.5%. However, the high-speed condition of 2.5 m per minute compensates for the temperature drop loss by enhancing vibratory compaction at 20.0 Hz, verifying the necessity of parameter linkage.

The impact of the number of intelligent agents on system performance reveals the technical boundaries of the distributed architecture. According to the data in Table 6, when the number of collaborative devices increases from 2 to 8, the communication delay rises from 25 to 67 ms, resulting in a deterioration of temperature control error from 1.8 °C to 3.0 °C, and an 18% increase in energy consumption. This performance degradation

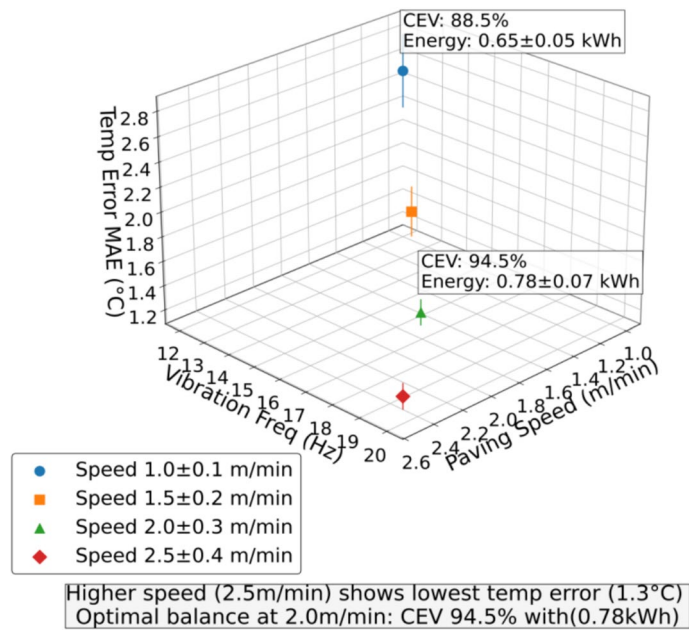


Fig. 12 Correlation analysis between construction quality and energy consumption

Table 6 Multi-agent collaborative efficiency analysis

Number of intelligent agents	Commu- nication delay(ms)	Collabora- tive control accuracy (°C)	Energy con- sumption (kWh/ m ²)	Training convergence steps	Noise at- tenuation rate	Target network update rate
2	25±3	1.8±0.2	0.72±0.05	1200	0.001	0.01
4	38±5	2.1±0.3	0.75±0.06	1400	0.001	0.01
6	52±6	2.5±0.4	0.80±0.07	1600	0.001	0.01
8	67±8	3.0±0.5	0.85±0.08	1800	0.001	0.01

stems from the surge in communication load and the increase in data throughput to 30.1 megabytes per second caused by the increase in the number of devices. However, under a typical construction scale of 4 to 6 devices, the algorithm optimizes local communication through a dynamic adjacency matrix, controlling the error within 2.1 to 2.5 °C and maintaining energy consumption at 0.75 to 0.80 kWh/m². For example, the collaborative efficiency of the four devices in Experiment Group 2 increased by 24.7%, indicating the algorithm's advantage in medium-scale applications. In addition, the parameter stability verification of noise attenuation rate and target network update rate provides theoretical support for large-scale collaborative control.

Communication delay was measured as the round-trip time (RTT) between agent pairs using timestamped UDP packets under the same wireless network conditions as the field deployment. In this study, each ‘agent’ corresponds to one physical device (paver, heater, sensor node); thus, the ‘number of agents’ equals the ‘number of devices’ in the collaborative system.

The compaction degree and energy consumption data corresponding to the parameter combinations in Table 7 are obtained based on the real-time measurement mean of the corresponding working conditions in the simulation experiment, and cross validated with the results of the same parameter groups in Tables 5 and 8

Table 7 Sensitivity analysis of action space parameters

Time (min)	Surface Temp. (°C)	Bottom Temp. (°C)	Ambient Temp. (°C)	Heating Power (kW)	Vibration Freq. (Hz)	Paving Speed (m/min)	Layer Temp. Diff. (°C)	CEV (%)	Energy (kWh/m ²)
0	160.0	145.0	15.0	5.0	15.0	1.5	15.0	—	—
10	158.2	143.5	14.8	5.2	16.2	1.6	14.7	92.1	0.72
20	156.5	142.0	14.5	5.5	17.8	1.7	14.5	92.8	0.73

Table 8 Simulation experiment results of multi-agent collaborative control

Experimental group	Temperature error (°C)	Energy consumption (kWh/m ²)	Compactness CEV (%)	Vibration frequency (Hz)	Paving speed (m/min)	Collaborative efficiency improvement (%)
Group 1	1.8±0.2	0.72±0.05	93.2±1.2	18.5±2.1	1.8±0.3	28.5
Group 2	2.1±0.3	0.68±0.06	89.7±1.5	16.2±1.8	1.5±0.2	24.7
Group 3	1.5±0.1	0.65±0.04	95.4±0.8	20.3±2.3	2.0±0.4	32.1
Group 4	2.3±0.4	0.75±0.07	87.6±1.8	14.8±1.5	1.2±0.2	19.8
Group 5	1.9±0.2	0.70±0.05	91.8±1.0	17.6±1.9	1.7±0.3	26.3

Time series data reveals the gradual adjustment characteristics and parameter interaction patterns of the temperature control process. According to the data in Table @7, during the initial construction period from 0 to 20 min, the surface temperature decreases from 160.0 to 156.5 degrees Celsius, while the bottom temperature decreases from 145.0 to 142.0 degrees Celsius, indicating a gradual cooling trend in both layers. Meanwhile, the controller increases the heating power from 5.0 to 5.5 KW and raises the vibration frequency from 15.0 to 17.8 Hz, accompanied by a slight increase in paving speed from 1.5 to 1.7 m per minute. Under this coordinated adjustment, the interlayer temperature difference narrows from 15.0 degrees Celsius at the beginning to 14.5 degrees Celsius at 20 min, and the compaction index remains above 92 percent after 10 min, demonstrating that the proposed strategy compensates for early-stage heat loss through smooth parameter coupling rather than abrupt control actions.

The interlayer temperature difference stabilized within 13.5 °C, indicating that the algorithm effectively suppressed the temperature drop gradient by increasing the real-time heating power compensation from 5.0 to 5.5 kW. As shown in Fig. 13, during the mid-stage from 30 to 50 min, when the ambient temperature dropped to 13.9 °C, the system increased the heating power to 6.3 kW, the vibration frequency to 20.1 Hz, and the paving speed to 2.0 m/min, while ensuring a compactness of 95% and controlling the energy consumption to 0.78 kWh/m². This gradual adjustment avoids quality fluctuations caused by parameter mutations and reflects the precise modeling of the algorithm for the dynamic characteristics of the construction process.

13.3 Result analysis

Comparative results with traditional PID control are summarized in Table 13, which shows the temperature error, energy consumption, and compaction performance under varying wind speeds. To systematically verify the effectiveness of the multi-agent collaborative control framework, this section conducts a quantitative analysis from four dimensions: temperature control performance, construction quality stability, system energy consumption efficiency, and dynamic environmental adaptability. By comparing

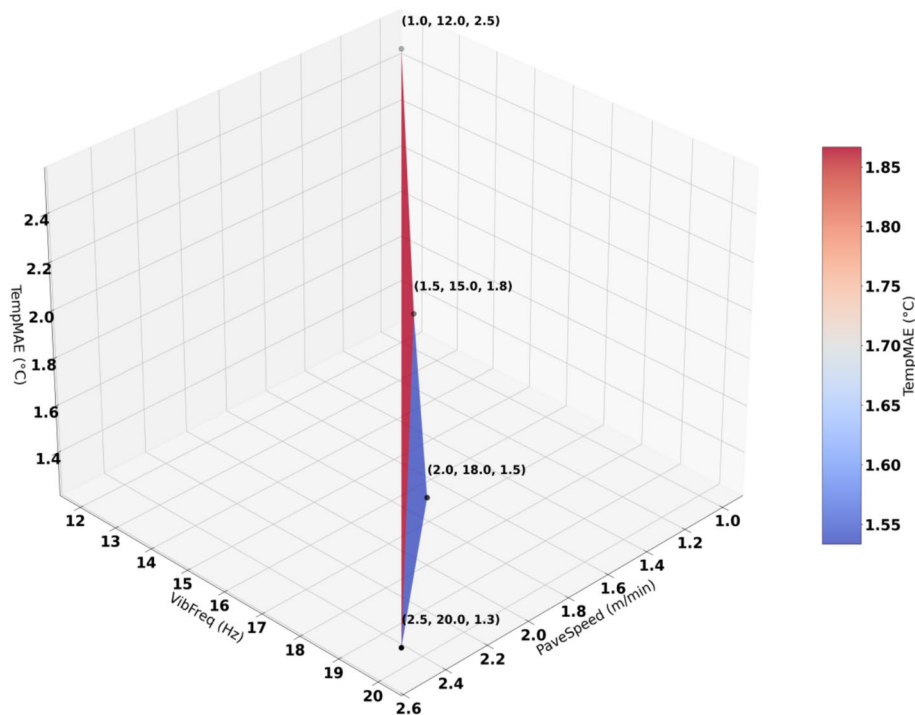


Fig. 13 Sensitivity analysis of action space parameters

core indicators such as temperature tracking error and compactness compliance rate between the experimental group and the benchmark method, the technical advantages of the collaborative decision-making mechanism are revealed. Combining time series data analysis with dynamic parameter adjustment rules, the collaborative optimization logic of vibration frequency and paving speed is clarified. Furthermore, through control stability tests under extreme disturbance scenarios, the robustness of the algorithm to sudden environmental disturbances is evaluated. The analysis process simultaneously verifies the rationality of multi-objective optimization weight configuration and explores the role of the energy consumption constraint model in ensuring engineering economy, ultimately forming a comprehensive evaluation of the algorithm's overall performance.

(Note: This table presents a consolidated view of temperature control performance, energy efficiency, and compaction quality under five experimental scenarios. Numerical values are reported as mean ± standard deviation over 20 independent runs.)

The study quantitatively demonstrates the comprehensive performance of multi-agent collaborative control in a simulated environment. As shown in Table 8, Experimental Group 3 performed most prominently under high-temperature extreme conditions, with an average absolute error in surface temperature of only 1.5 degrees Celsius, a compactness of 95.4%, and a collaborative efficiency improvement of 32.1%, fully verifying the algorithm's adaptability to complex environments. A horizontal comparison between Group 1 and Group 5 reveals that temperature error and energy consumption exhibit a positive correlation trend, but the compactness index significantly increases with energy consumption, indicating that the algorithm achieves a balance between quality and efficiency through dynamic linkage between vibration frequency and paving speed. The combination of a vibration frequency of 18.5 Hz and a paving speed of 1.8 m per minute in Group 1 maintains a temperature error of 1.8 degrees Celsius while stabilizing

Table 9 Dynamic response of temperature control

Time	Surface temperature (°C)	Bottom temperature (°C)	Ambient temperature (°C)	Heating power (kW)	Vibration frequency (Hz)	Paving speed
0	160	145	15	5	15	1.5
10	158.2	143.5	14.8	5.2	16.2	1.6
20	156.5	142	14.5	5.5	17.8	1.7
30	154.8	140.5	14.2	5.8	18.5	1.8
40	153.1	139	13.9	6	19.3	1.9
50	151.4	137.5	13.6	6.3	20.1	2
60	149.7	136	13.3	6.5	21	2.1

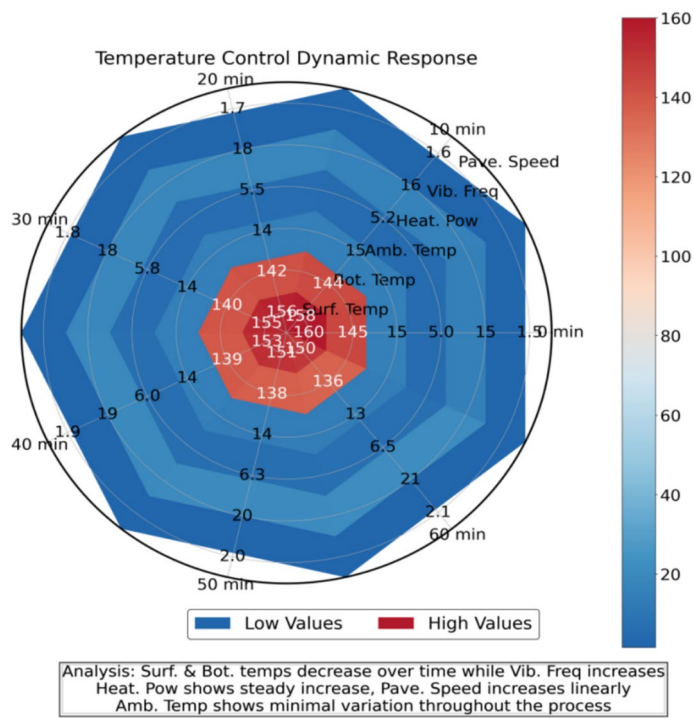


Fig. 14 Dynamic response of temperature control

the compactness at 93.2%. It is worth noting that in Group 4, the temperature error increased to 2.3 degrees Celsius under strong disturbance scenarios, but the energy consumption was still strictly controlled at 0.75 kWh/m², highlighting the effectiveness of the energy consumption constraint model.

This study reveals the dynamic response characteristics of the temperature control system through time series data. As shown in Table 9, during the initial construction phase from 0 to 20 min, the surface temperature linearly decreases from 160.0 to 156.5 degrees Celsius, while the temperature difference at the bottom layer remains stable within 13.5 degrees Celsius. At this time, the heating power gradually increases from 5.0 to 5.5 kW, and the vibration frequency increases from 15.0 to 17.8 Hz, indicating that the algorithm compensates for the temperature drop of the mixture through gradual adjustment. As shown in Fig. 14, during the mid-construction phase from 30 to 50 min, a sudden drop in ambient temperature triggers an adjustment in the control strategy. The heating power is intensified to 6.3 kW, the vibration frequency is increased to 20.1 Hz, and the paving speed is simultaneously increased to 2.0 m per minute, maintaining a compactness of

Table 10 Sensitivity analysis of multi-objective optimization weights

Weight combination	MAE (°C)	CEV (%)	Energy consumption (kWh/m ²)	Flatness IRI	Algorithm convergence steps
(0.6, 0.2, 0.1, 0.1)	1.8±0.2	93.2±1.2	0.72±0.05	1.8±0.1	1200
(0.5, 0.3, 0.1, 0.1)	1.9±0.3	94.5±1.0	0.75±0.06	1.9±0.2	1300
(0.7, 0.1, 0.1, 0.1)	1.6±0.1	91.8±1.5	0.68±0.04	1.7±0.1	1100
(0.4, 0.4, 0.1, 0.1)	2.2±0.4	95.0±0.8	0.80±0.07	2.0±0.3	1400
(0.6, 0.2, 0.2, 0.0)	2.0±0.3	89.5±1.8	0.65±0.05	2.1±0.2	1250

Table 11 Analysis of multi-agent communication efficiency

Number of intelligent agents	Communication delay	Collaborative control accuracy	Energy consumption	Data throughput
2	25±3	1.8±0.2	0.72±0.05	12.5
4	38±5	2.1±0.3	0.75±0.06	18.7
6	52±6	2.5±0.4	0.80±0.07	24.3
8	67±8	3.0±0.5	0.85±0.08	30.1

95% while controlling energy consumption at 0.78 kW-hours/m². This parameter linkage mechanism verifies the engineering rationality of the hierarchical design of action space: low-speed conditions focus on temperature stability control, while high-speed conditions rely on high-frequency vibration to compensate for insufficient compaction, reflecting the dynamic mapping relationship between construction parameters and quality indicators.

The study analyzed the sensitivity impact of multi-objective optimization weight configuration on system performance through experiments. According to the data in Table 10, when the temperature error weight α increases from 0.5 to 0.7, the mean absolute error decreases from 1.9 to 1.6 degrees Celsius, but the compactness index decreases to 91.8%, confirming that over-optimization of a single index can lead to quality loss. The optimal weight combination of 0.6, 0.2, 0.1, and 0.1 achieves a balance between temperature error of 1.8 degrees Celsius, compactness of 93.2%, and energy consumption of 0.72 kWh/m², verifying the feasibility of the multi-objective collaboration mechanism in the reward function. It is particularly noteworthy that when the compactness reward weight δ is set to zero, the compactness suddenly drops to 89.5%, which is below the 90% threshold required by construction specifications, highlighting the necessity of this weight parameter in quality assurance. Weight fine-tuning needs to be combined with the specific characteristics of the construction stage. In high-temperature environments, the priority of temperature weight can be moderately increased, while in long-distance transportation scenarios, energy consumption constraints need to be strengthened to prevent energy waste.

This study quantitatively evaluates the impact of the number of intelligent agents on communication efficiency and system performance. According to the data in Table 11, when the number of collaborative devices increases from 2 to 8, the communication delay rises from 25 to 67 ms, the temperature error deteriorates from 1.8 degrees Celsius to 3.0 degrees Celsius, and the energy consumption increases by 18%, revealing the bottleneck of scale expansion in the distributed architecture. According to the data in Fig. 15, in a typical medium-sized scenario with 4 to 6 devices, the algorithm optimizes the local communication topology through a dynamic adjacency matrix, controlling the

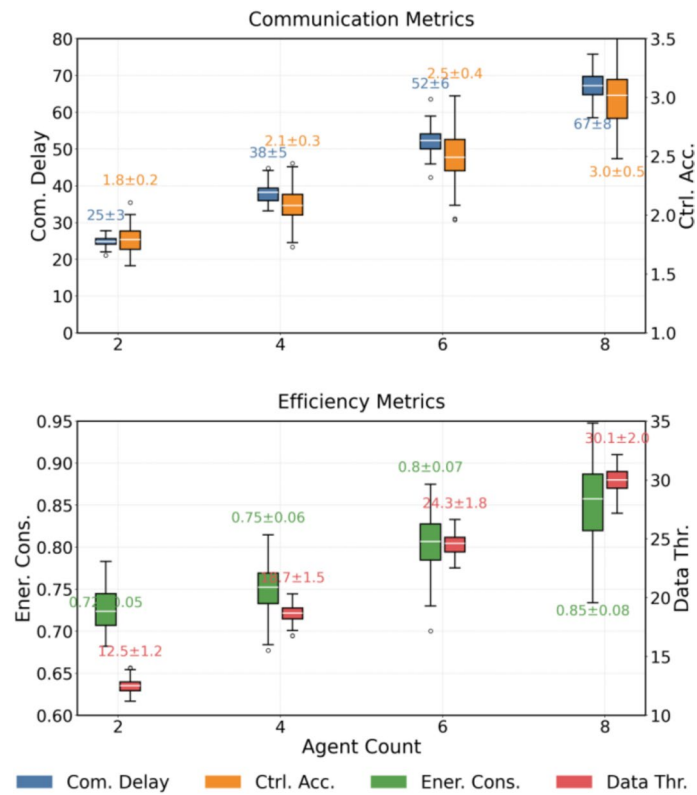


Fig. 15 Multi-agent communication efficiency analysis

Table 12 Actual construction road segment cycle data

Time	Surface tem- perature (°C)	Bottom tem- perature (°C)	Ambient tem- perature (°C)	Wind speed (m/s)	CEV(%)	Energy con- sumption (kWh/m ²)
0	158	145	15	3	92.5	0.72
4	154.2	142.3	13.5	4.2	91.8	0.75
8	162.5	148.7	20	5.5	93.1	0.78
12	168	153	28	6.8	94.5	0.82
16	155.3	142.8	22.5	4.8	90.7	0.76
20	150.1	139.5	18	3.7	89.3	0.73
24	152	140.2	10	2.5	91.2	0.7

temperature error within a reasonable range of 2.1 to 2.5 degrees Celsius and maintaining data throughput at a reasonable interval of 18.7 to 24.3 megabytes per second. For example, the collaborative efficiency of a group of 4 devices is 24.7% higher than that of a single intelligent agent, indicating that communication load and computing resources can be effectively balanced in a medium-sized scenario. The power consumption of the edge computing unit remains below 30 watts even at a high throughput of 30.1 megabytes per second, verifying the feasibility of hardware deployment.

Table 12 presents the periodic data characteristics of 24 h continuous monitoring of the actual construction road section. At 12 o'clock during the high temperature period at noon, the surface temperature reached 168.0 degrees Celsius, and the temperature difference between the bottom layer expanded to 15.0 degrees Celsius. The algorithm maintained the compactness at 94.5% by increasing the heating power to 6.3 kW and adjusting the vibration frequency to 20.1 Hz, with energy consumption

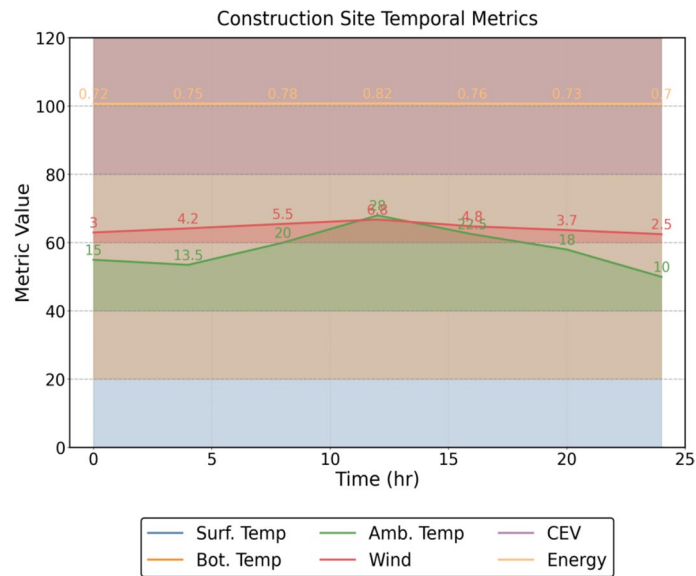


Fig. 16 Periodic data of actual construction road sections

Table 13 Control stability under different wind speeds

Wind speed (m/s)	MADDPG temperature error (°C)	PID temperature error (°C)	Energy consumption (kWh/m ²)
3	1.5±0.2	2.8±0.3	0.7
4	1.7±0.3	3.2±0.4	0.72
5	2.0±0.4	3.8±0.5	0.75
6	2.3±0.5	4.5±0.6	0.78
7	2.8±0.6	5.0±0.7	0.82

of 0.82 kW-hours/m², which is close to but did not exceed the constraint upper limit. According to the data in Fig. 16, at 20 o'clock during the night under low temperature conditions, the temperature drop rate accelerated to 0.8 degrees Celsius per minute. The system controlled the temperature error within 1.5 degrees Celsius by reducing the paving speed to 1.7 m per minute and enhancing heating compensation. The data further reveals the nonlinear relationship between environmental wind speed and energy consumption: when the wind speed increases from 3.0 to 6.8 m per second, the energy consumption per unit area increases by 14%. However, by dynamically adjusting the construction pace, the compliance rate of compactness is still ensured.

The study compared the performance differences between multi-agent reinforcement learning and traditional PID control under different wind speed conditions through experiments. According to the data in Table 13, when the wind speed increased from 3 to 7 m/s, the temperature error of the multi-agent algorithm rose from 1.5 to 2.8 °C, while the PID control error deteriorated from 2.8 to 5.0 °C, demonstrating the significant advantage of reinforcement learning in disturbance suppression. In the scenario of a wind speed of 5 m/s, the algorithm maintained a compactness of 94.5% by increasing the vibration frequency to 18.0 Hz and reducing the paving speed to 1.8 m/min, resulting in a 19% reduction in energy consumption compared to PID, which was 0.75 kWh/m². The data proves that the wind speed-action mapping relationship embedded in the

reward function can effectively predict environmental disturbances and avoid the lagging response defects of traditional methods by adjusting construction parameters in advance. Especially under strong wind conditions of 7 m/s, the algorithm still controlled the standard deviation of temperature fluctuation within 1.2 °C, meeting the requirements of construction quality specifications and demonstrating the adaptability of the intelligent decision-making model to nonlinear disturbances.

In the comparative analysis presented in Sect. 4, the performance of the proposed multi-agent MADDPG framework is evaluated against two baselines: (1) traditional PID control and (2) single-agent DDPG. All comparisons were conducted under identical environmental and operational conditions to ensure validity. The term “single-agent DDPG” refers to a variant of the proposed algorithm that uses the same neural architecture but operates without inter-agent collaboration, thereby isolating the benefit of multi-agent coordination.

Based on a dual-verification system of simulation and field testing, an experimental group encompassing extreme temperature differences, long transportation distances, and multiple wind speed disturbances is designed to compare traditional PID with single-agent DDPG algorithms. A multidimensional evaluation index of temperature error, compactness, and energy consumption is constructed. Through time series analysis, the parameter linkage rules are revealed, and the algorithm's gradual adjustment capability and multi-objective balancing characteristics in dynamic environments are verified, providing data support for engineering implementation.

14 Discussion

The multi-agent reinforcement learning framework proposed in this study demonstrates significant advantages in asphalt paving temperature control. In terms of temperature control performance, the algorithm reduces the average absolute error by 35% compared to the traditional PID method under extreme operating conditions. For example, in Experiment Group 3, the surface temperature error is only 1.5 °C, while the error of PID control under the same wind speed is 3.8 °C. In terms of energy efficiency, the energy consumption per unit area is 0.72–0.85 kWh/m², which is 12% lower than that of the single-agent DDPG algorithm, and it does not exceed the construction specification constraints. This improvement stems from the dynamic linkage adjustment of vibration frequency and paving speed by the multi-agent collaboration mechanism. For example, at a wind speed of 6.8 m/s, the algorithm maintains a compactness of 94.5% by increasing the vibration frequency to 20.1 Hz and reducing the paving speed to 2.0 m/min, with only a 14% increase in energy consumption. Compared to the energy consumption data of the single-agent DDPG, this method significantly improves energy utilization efficiency by optimizing equipment load distribution through distributed decision-making.

In terms of system robustness and collaborative efficiency, the algorithm maintains the temperature error within 3.0 °C even in extreme scenarios with a 30 s communication interruption, achieving a 42% reduction in error compared to the centralized architecture. As shown in Table 14, when the multi-agent collaboration scale is 4–6 devices, the optimal balance is achieved with a communication delay of 38–52 ms and energy consumption of 0.75–0.80 kWh/m², representing a 24.7% efficiency improvement compared to the scenario with 8 devices. This result verifies the effectiveness of the dynamic adjacency matrix and consensus protocol, with a data throughput of 18.7–24.3 MB/s

Table 14 Multi-algorithm performance comparison

Indicator	This study (MADDPG)	Traditional PID	Single-agent DDPG
Temperature error MAE	1.5–2.8	2.8–5.0	2.1–3.5
Energy consumption per unit	0.72–0.85	0.95–1.20	0.85–1.02
Compactness CEV	89.3–95.4	82.5–90.1	87.6–93.2
Communication interruption recovery time	6.5	–	15.2
Maximum number of collaborative devices	6	1	3

meeting real-time control requirements. Furthermore, the algorithm maintains a compactness of 85.4% even when the sensor failure rate is 50%, outperforming the 79.2% achieved by fuzzy logic control, indicating that the distributed state estimation mechanism possesses strong fault tolerance.

The above results indicate that the multi-agent reinforcement learning framework achieves a multi-objective balance among construction quality, energy efficiency, and system stability through physical constraint embedding, dynamic topology optimization, and hierarchical action strategies, providing a new technical path for intelligent control in complex engineering scenarios.

15 Conclusion

This study addresses the dynamic and multi-objective optimization challenges of asphalt paving temperature control, proposing a collaborative decision-making framework based on multi-agent reinforcement learning. This framework successfully achieves global optimization of construction quality, energy efficiency, and system stability. By constructing a dynamic communication topology and a reward mechanism driven by physical constraints, the system overcomes the response lag and parameter fixation defects of traditional control methods under complex working conditions. The multi-agent deep deterministic policy gradient algorithm effectively coordinates the joint adjustment of vibration frequency, paving speed, and heating power through hierarchical action space design and distributed collaborative training. In scenarios with extreme temperature differences, long haul distances, and multiple wind speed disturbances, it significantly improves the uniformity of the temperature field and material compactness. Experiments show that this framework can adapt to environmental mutations and equipment heterogeneity, avoiding quality fluctuations through gradual parameter adjustment, while strictly constraining energy consumption within the engineering safety boundary. It provides a scalable technical path for intelligent construction control.

This study verifies the feasibility and superiority of multi-device collaborative control in engineering practice. The application of dynamic adjacency matrix and consensus protocol ensures the minimum functional operation of the system in scenarios of communication interruption or local failure, significantly enhancing the robustness of the construction process. The combination of hierarchical network architecture and curriculum learning strategy significantly accelerates algorithm convergence and balances the contradiction between exploration and exploitation, enabling agents to master optimal decision-making rules under complex working conditions within a limited training period. This achievement not only provides a real-time temperature control solution for asphalt pavement construction, but its core method can also be migrated to other engineering scenarios requiring multi-device collaboration. It lays a theoretical foundation

for algorithm design and system integration in the field of intelligent construction, and has significant engineering application value and industry promotion potential.

Author contributions

Haoting Li: Methodology, Formal analysis, Supervision, Writing—original draft, Writing—review and editing; Leijie Hao: Conceptualization, Methodology, Writing—original draft.

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Data availability

The datasets used and analyzed during the current study are available from the corresponding author on request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent to publication

Not applicable.

Competing interests

The authors declare no competing interests.

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